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NITRIDE-BASED SEMICONDUCTOR CIRCUIT AND METHOD FOR MANUFACTURING THE SAME

Abstract

A nitride-based semiconductor circuit including a substrate structure, a nitride-based heterostructure, connectors, and connecting vias is provided. The substrate structure includes a first type semiconductor substrate, and a second type semiconductor substrate. The second type semiconductor substrate is embedded in a region of the first type semiconductor substrate. The first type semiconductor substrate has first dopants, and the second type semiconductor substrate has second dopants to form a pn junction between the first type semiconductor substrate and the second type semiconductor substrate. The nitride-based heterostructure is disposed on the substrate structure. The connectors are disposed on the nitride-based heterostructure. The connecting vias include a first interconnection and a second interconnection. The first interconnection electrically connects the first region of the first type semiconductor substrate to one of the connectors. The second interconnection electrically connects the second type semiconductor substrate to another one of the connectors.

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Background/Summary

FIELD OF THE DISCLOSURE
[0001] The present disclosure generally relates to a nitride-based semiconductor circuit. More specifically, the present disclosure relates to a nitride-based semiconductor circuit having a pn junction formed in its substrate.
BACKGROUND
[0002] In recent years, intense research on high-electron-mobility transistors (HEMTs) has been prevalent, particularly for high power switching and high frequency applications. III-nitride-based HEMTs utilize a heterojunction interface between two materials with different bandgaps to form a quantum well-like structure, which accommodates a two-dimensional electron gas (2DEG) region, satisfying demands of high power/frequency devices. In addition to HEMTs, examples of devices having heterostructures further include heterojunction bipolar transistors (HBT), heterojunction field effect transistors (HFET), and modulation-doped FETs (MODFET).
[0003] Due to characteristics of gallium nitride (GaN), GaN-based devices can be applied to half bridge circuits. Half bridge circuits require a high side transistor and a low side transistor. The transistors need to be alternately switched on. However, while one of the transistors is switched on, another transistor will be affected by voltage leakage. Therefore, electronic circuits having transistors that can be switched on without affecting other transistors are needed.
SUMMARY OF THE DISCLOSURE
[0004] In accordance with one aspect of the present disclosure, a nitride-based semiconductor circuit is provided. The nitride-based semiconductor circuit includes a substrate structure, a nitride-based heterostructure, a plurality of connectors, and a plurality of connecting vias. The substrate structure includes a first type semiconductor substrate, and a second type semiconductor substrate. The first type semiconductor substrate has a first region and a second region, and the second region is adjacent to the first region. The second type semiconductor substrate is embedded in the second region of the first type semiconductor substrate. A top surface of the first type semiconductor substrate and a top surface of the second type semiconductor substrate are substantially coplanar. The first type semiconductor substrate has first dopants, and the second type semiconductor substrate has second dopants to form a pn junction between the first type semiconductor substrate and the second type semiconductor substrate. The nitride-based heterostructure is disposed on the substrate structure. The connectors are disposed on the nitride-based heterostructure. The connecting vias go through the nitride-based heterostructure. The connecting vias include a first interconnection and a second interconnection. The first interconnection electrically connects the first region of the first type semiconductor substrate to one of the connectors. The second interconnection electrically connects the second type semiconductor substrate to another one of the connectors.

[0005] In accordance with one aspect of the present disclosure, a nitride-based semiconductor circuit is provided. The nitride-based semiconductor circuit includes a first type semiconductor substrate, a second type semiconductor substrate, a nitride-based heterostructure, a first drain connector, a first source connector, a first gate connector, a second drain connector, a second source connector, and a second gate connector. The first type semiconductor substrate has a first region and a second region adjacent to the first region. The second type semiconductor substrate is embedded in the second region of the first type semiconductor substrate. A top surface of the first type semiconductor substrate and a top surface of the second type semiconductor substrate are coplanar. The first type semiconductor substrate has first dopants, and the second type semiconductor substrate has second dopants to form a pn junction between the first type semiconductor substrate and the second type semiconductor substrate. The nitride-based heterostructure is disposed on both the first type semiconductor substrate and the second type semiconductor substrate. The first gate connector is disposed between the first drain connector and the first source connector. The second gate connector is disposed between the second drain connector and the second source connector. The first source connector shares the same voltage with one of the semiconductor substrates. The second source connector shares the same voltage with another one of the semiconductor substrates. Voltages of the first and second type semiconductor substrates reversibly bias the pn junction therebetween.
[0006] In accordance with one aspect of the present disclosure, a manufacturing method of a nitride-based semiconductor circuit is provided. The manufacturing method includes: providing a substrate structure, wherein the substrate structure comprises a first type semiconductor substrate having first dopants, and the first type semiconductor substrate has a first region and a second region adjacent to the first region; and a second type semiconductor substrate embedded in the second region of the first type semiconductor substrate, wherein a top surface of the first type semiconductor substrate and a top surface of the second type semiconductor substrate are coplanar, and the first type semiconductor substrate has first dopants, and the second type semiconductor substrate has second dopants to form a pn junction between the first type semiconductor substrate and the second type semiconductor substrate; forming a nitride-based heterostructure on the substrate structure; forming a plurality of connectors on the nitride-based heterostructure; etching through the nitride-based heterostructure; and forming a plurality of connecting vias. The connecting vias include a first interconnection and a second interconnection. The first interconnection electrically connects the first region of the first type semiconductor substrate to one of the connectors. The second interconnection **191** electrically connects the second type semiconductor substrate to another one of the connectors.
[0007] By the above configuration, the nitride-based semiconductor circuit has a substrate structure with a pn junction, and voltage can be applied through the connectors, so as to bias the pn junction of the substrate structure, and voltage leakage may be prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] Aspects of the present disclosure are readily understood from the following detailed description when read with the accompanying figures. It should be noted that various features may not be drawn to scale. That is, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion. Embodiments of the present disclosure are described in more detail hereinafter with reference to the drawings, in which:
[0009] FIG. **1** is a top view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0010] FIG. **2** is a cross-sectional view of the nitride-based semiconductor circuit according to

cutting plane line I**1** in FIG. **1**;
[0011] FIGS. **3-16** are cross-sectional views of steps of a manufacturing method of the nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0012] FIG. **17** is a top view of a nitride-based semiconductor circuit according to another embodiment of the present disclosure;
[0013] FIG. **18** is a cross-sectional view of a semiconductor circuit according to cutting plane line I**2** in FIG. **17**;
[0014] FIG. **19** is a top view of a nitride-based semiconductor circuit according to another embodiment of the present disclosure;
[0015] FIG. **20** is a cross-sectional view of the nitride-based semiconductor circuit according to cutting plane line I**3** in FIG. **19**;
[0016] FIG. **21** is a cross-sectional view of a nitride-based semiconductor circuit according to another embodiment of the present disclosure;
[0017] FIG. **22** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0018] FIG. **23** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0019] FIG. **24** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0020] FIGS. **25-28** are cross-sectional views of steps of manufacturing the nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0021] FIG. **29** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0022] FIG. **30** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0023] FIG. **31** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0024] FIG. **32** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0025] FIG. **33** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0026] FIG. **34** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0027] FIG. **35** is a top view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0028] FIG. **36** is a cross-sectional view of the nitride-based semiconductor circuit according to cutting plane line I**4** in FIG. **35**;
[0029] FIGS. **37-45** are cross-sectional views of steps of a manufacturing method of the nitride-based semiconductor circuit;
[0030] FIG. **46** is a top view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0031] FIG. **47** is a cross-sectional view of the nitride-based semiconductor circuit according to cutting plane line I**5** in FIG. **46**;
[0032] FIG. **48** is a top view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure;
[0033] FIG. **49** is a cross-sectional view of the nitride-based semiconductor circuit according to cutting plane line I**6** in FIG. **48**; and
[0034] FIG. **50** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0035] Common reference numerals are used throughout the drawings and the detailed description to indicate the same or similar components. Embodiments of the present disclosure will be readily understood from the following detailed description taken in conjunction with the accompanying drawings.
[0036] Spatial descriptions, such as “on,” “above,” “below,” “up,” “left,” “right,” “down,” “top,” “bottom,” “vertical,” “horizontal,” “side,” “higher,” “lower,” “upper,” “over,” “under,” and so forth, are specified with respect to a certain component or group of components, or a certain plane of a component or group of components, for the orientation of the component(s) as shown in the associated figure. It should be understood that the spatial descriptions used herein are for purposes of illustration only, and that practical implementations of the structures described herein can be spatially arranged in any orientation or manner, provided that the merits of embodiments of this disclosure are not deviated from by such arrangement.
[0037] Further, it is noted that the actual shapes of the various structures depicted as approximately rectangular may, in actual device, be curved, have rounded edges, have somewhat uneven thicknesses, etc. due to device fabrication conditions. The straight lines and right angles are used solely for convenience of representation of layers and features.
[0038] In the following description, semiconductor circuits/devices/dies/packages, methods for manufacturing the same, and the likes are set forth as preferred examples. It will be apparent to those skilled in the art that modifications, including additions and/or substitutions may be made without departing from the scope and spirit of the present disclosure. Specific details may be omitted so as not to obscure the present disclosure; however, the disclosure is written to enable one skilled in the art to practice the teachings herein without undue experimentation.
[0039] FIG. **1** is a top view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure, and FIG. **2** is a cross-sectional view of the nitride-based semiconductor circuit according to cutting plane line I**1** in FIG. **1**, and parts of the layers of the nitride-based semiconductor circuit are omitted in the top view for clarity. The nitride-based semiconductor circuit **1**A comprises a semiconductor substrate **12**, a semiconductor substrate **13**, and a nitride-based heterostructure **14**. The substrate **13** is disposed on the substrate **12**, and the nitride-based heterostructure **14** is disposed on the substrate **13**.
[0040] In this embodiment, the substrate **12** has dopants, and the substrate **13** has dopants. The dopants of the substrate **12** are different from the dopants of the substrate **13**, and a pn junction **121** is formed between the substrates **12**, **13**. In other words, the substrate **12** and the substrate **13** forms an interface therebetween with pn junction **121** across the interface.
[0041] For example, the semiconductor substrates **12**, **13** may include silicon (Si).
[0042] The nitride-based semiconductor circuit **1**A further comprises a plurality of connectors **15**, a patterned conductive layer **20**, a patterned conductive layer **21**, and a plurality of connecting vias **19**. The connectors **15** are disposed on the nitride-based heterostructure **14**, and the patterned conductive layers **20**, **21** are disposed on the connectors **15**.
[0043] The connecting vias **19** go through the nitride-based heterostructure **14**. To be specific, a plurality of holes go through the nitride-based heterostructure **14**, and a plurality of connections are formed between the layer above the nitride-based heterostructure **14** and the layer below the nitride-based heterostructure **14** through the holes.
[0044] In this embodiment, the connecting vias **19** comprise: interconnection **190** and interconnection **191**. The interconnection **190** electrically connects the substrate **12** to one of the connectors **15** through the patterned conductive layer **21**. The interconnection **191** electrically connects the substrate **13** to another one of the connectors **15** through the patterned conductive layer **21**. Therefore, the voltage of the substrate **12** and the voltage of the substrate **13** can be controlled through applying electrical signals to the connectors **15**, and the pn junction **121** may be biased.
[0045] The interconnection **190** separates a group of connectors **15** from another group of

connectors **15**. When the pn junction **121** is zero biased or reverse biased, no voltage leakage may be allowed between the groups of connectors **15**.

[0046] In some embodiments, the connectors **15** and connecting vias **19** can include, for example but are not limited to, metals, alloys, doped semiconductor materials (such as doped crystalline silicon), compounds such as silicides and nitrides, other conductor materials, or combinations thereof. The exemplary materials of the connectors **15** and connecting vias **19** can include, for example but are not limited to, W, Au, Pd, Ta, Co, Ni, Pt, Mo, Ti, AlSi, TiN, or combination thereof. The connectors **15** and connecting vias **19** may be a single layer, or plural layers of the same or different composition. In some embodiments, the connectors **15** and connecting vias **19** form ohmic contacts with the nitride-based heterostructure **14** or substrates **12**, **13**. The ohmic contacts can be achieved by applying Ti, Al, or other suitable materials to the connectors **15** and connecting vias **19**. In some embodiments, each of the connectors **15** and connecting vias **19** is formed by at least one conformal layer and a conductive filling. The conformal layer can wrap the conductive filling. The exemplary materials of the conformal layer, for example but are not limited to, Ti, Ta, TiN, Al, Au, AlSi, Ni, Pt, or combinations thereof. The exemplary materials of the conductive filling can include, for example but are not limited to, AlSi, AlCu, or combinations thereof.

[0047] Referring to FIG. **2**, the connectors **15** comprise source connectors **151**, **153**, drain connectors **150**, **154**, and gate connectors **152**, **155**. The gate connector **152** is located between the drain connector **150** and the source connector **151**, and the gate connector **155** is located between the drain connector **154** and the source connector **153**.

[0048] The source connector **151**, the drain connector **150**, the gate connector **152**, and a part of the nitride-based heterostructure **14** are configured to constitute a high-electron-mobility transistor (HEMT) structure **1***a*. The source connector **153**, the drain connector **154**, the gate connector **155**, and another part of the nitride-based heterostructure **14** are configured to constitute another HEMT structure **1***b*. The HEMT structure **1***a* is located beside the HEMT structure **1***b*. The HEMT structure **1***a* and the HEMT structure **1***b* are adjacent.

[0049] In this embodiment, a threshold voltage of the HEMT structure **1***b* is different from a threshold voltage of the HEMT structure **1***a*. To be specific, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*. Therefore, the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally being turned on by a voltage spike or voltage leakage.

[0050] Also, the HEMT structure **1***a* is connected to the HEMT structure **1***b*. To be specific, the patterned conductive layer **21** connects the source connector **151** to the drain connector **154**, optionally the HEMT structures **1***a* and **1***b* may form a half bridge circuit.

[0051] The threshold voltage of the HEMT structure **1***b* is higher than the HEMT structure **1***a*, and the HEMT structure **1***b* can be prevented from being affected by the voltage of the HEMT structure **1***a* during operation. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally being opened by a voltage spike or voltage leakage.

[0052] Moreover, the gate connector **152** has a gate electrode **1521** and a doped nitride-based semiconductor layer **1522**, and the gate connector **155** has a gate electrode **1551** and a doped nitride-based semiconductor layer **1552**. The doped nitride-based semiconductor layers **1522**, **1552** are disposed on the nitride-based heterostructure **14**, and the gate electrode **1521** is disposed on the doped nitride-based semiconductor layer **1522**, and the gate electrode **1551** is disposed on the doped nitride-based semiconductor layer **1552**. Therefore, the HEMT structures **1***a*, **1***b* are enhanced mode HEMTs. For example, the doped nitride-based semiconductor layers **1522**, **1552** are p-type doped gallium nitride layers. With the higher threshold voltage, the HEMT structure **1***b* can be prevented from accidentally turning on while the voltage of the source connector **151** is changing. The HEMT structure **1***a* can be prevented from being opened while the HEMT structure

1*b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally being turned on by a voltage spike or a voltage leakage.

[0053] In one aspect, the interconnection **190** electrically connects the drain connector **150** to the substrate **12**, and the interconnection **191** electrically connects the source connectors **151**, **153** to the substrate **13**. The pn junction **121** can be biased through the interconnections **190**, **191**.

[0054] For example, the substrate **12** has n-type dopants, and the substrate **13** has p-type dopants. The substrate **12** is connected to the drain connector **150** through the interconnection **190**, and the substrate **13** is connected to the source connector **151** and the source connector **153** through the interconnection **191**.

[0055] The substrate **13** shares the same voltage as the source connector **151** or the source connector **153**. The substrate **12** shares the same voltage as the drain connector **150**. Therefore, the n-type doped substrate **12** is connected to the drain of the HEMT structure **1***a* or a V.sub.dd of the nitride-based semiconductor circuit **1**A, and the p-type doped substrate **13** is connected to the sources of the HEMT structures **1***a* and **1***b*. The pn junction **121** is reversely biased by the voltages of the substrates **12**, **13**, and a depletion region is formed, and the substrate voltage of the HEMT structure **1***a* is isolated from the substrate voltage of the HEMT structure **1***b*. In other words, the substrate voltage of the HEMT structure **1***a* and the substrate voltage of the HEMT structure **1***b* may be different.

[0056] Moreover, the substrate **12** has a heavily doped area **120**, and the interconnection **190** is connected to the heavily doped area **120**.

[0057] FIG. **1** only shows a top view of the substrate **12**, the heavily doped area **120**, and part of the patterned conductive layer **21** thereon for clarity. Therefore, the substrate voltage of the HEMT structure **1***a* and the substrate voltage of the HEMT structure **1***b* can be properly separated through interconnection **190**.

[0058] Moreover, the heavily doped area **120** surrounds a projection **1***c* of the HEMT structure **1***a* on the substrate **12**, and a projection **1***d* of the HEMT structure **1***b* on the substrate **12** is not surrounded by the heavily doped area **120**. Therefore, the substrate voltage of the HEMT structure **1***a* is isolated from the substrate voltage of the HEMT structure **1***b*. In other words, the substrate **12** and the substrate **13** form a plurality of diode structures, and the diode structures surrounded by the heavily doped area **120** is isolated from the diode structures that is not surrounded by the heavily doped area **120**.

[0059] For example, when the HEMT structure **1***a* is on and the HEMT structure **1***b* is off, the pn junction **121** between the substrates **12**, **13** under the HEMT structure **1***a* is zero biased, and the pn junction **121** under the HEMT structure **1***b* is reversely biased, and a depletion region is created. The voltage of the substrate **13** under the HEMT structure **1***b* is isolated while the voltage of the substrate **13** under the HEMT structure **1***a* is changing.

[0060] When the HEMT structure **1***a* is off and the HEMT structure **1***b* is on, the pn junctions **121** under the HEMT structures **1***a*, **1***b* are both reversely biased, and depletion regions are created. The voltage of the substrate **13** under the HEMT structure **1***a* is isolated while the voltage of the substrate **13** under the HEMT structure **1***b* is changing.

[0061] In one aspect, the nitride-based semiconductor circuit **1**A comprises passivation layers **16**, and the gate connector **155** of the HEMT structure **1***b* is covered by multiple passivation layers **16**, while the gate connector **152** of the HEMT structure **1***a* is covered by single passivation layer.

[0062] To be specific, the nitride-based semiconductor circuit **1**A comprises an oxide layer **160** and a nitride layer **161**. The oxide layer **160** is in contact with the gate connector **155**, and the oxide layer **160** is isolated from the gate connector **152**. The oxide layer **160** abuts the gate connector **155**, and no oxide layer **160** makes contact with the gate connector **152** that is surrounded by the interconnection **190**.

[0063] In other words, the interconnection **190** is disposed on the heavily doped area **120**, and the interconnection **190** surrounds an area in the nitride-based semiconductor circuit **1**A. The drain

connector **150**, the source connector **151**, and the gate connector **152** are located in the area surrounded by the interconnection **190**, and the gate connector **152** is free from oxide layer **160**. The nitride layer **161** is in contact with the gate connector **152**, and no other passivation layer makes contact with gate connector **152**.
[0064] For example, in this embodiment, the oxide layer **160** comprises silicon dioxide, and the oxide layer **160** is applied through plasma-enhanced chemical vapor deposition (PECVD). The nitride layer **161** comprises silicon nitride, and the nitride layer **161** is applied through low pressure chemical vapor deposition (LPCVD). The hydrogen concentration of oxide layer **160** is different from the hydrogen concentration of nitride layer **161**, and the doped nitride-based semiconductor layer **1552** is in direct contact with the oxide layer **160**. While the doped nitride-based semiconductor layers **1522**, **1552** are doped with a p-type dopant, activation of magnesium in doped nitride-based semiconductor layer **1522** and activation of magnesium in doped nitride-based semiconductor layer **1552** can be different during high temperature thermal environment. Therefore, the threshold voltage of the HEMT structure **1***a* is different from the threshold voltage of the HEMT structure **1***b*. The HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from being accidentally turned on by a voltage spike or a leakage voltage.
[0065] In one aspect, the nitride-based semiconductor circuit **1**A further comprises a semiconductor substrate **11**. The substrate **12** is disposed on the substrate **11**, and the substrates **11**, **13** have the same type of dopants. For example, when the substrates **11**, **13** have p-type dopants, and the substrate **12** have n-type dopants, the substrates **11-13** together form a p-n-p structure. The substrate **11** and the substrate **12** also form a pn junction **111**.
[0066] To be specific, the bottom side of the interconnection **190** is located between the pn junction **111** and the pn junction **121**, and the bottom side of the interconnection **191** is located above the pn junction **121**. Therefore, the pn junction **121** can be reversely biased or zero biased through the interconnections **190**, **191**.
[0067] In one aspect, the nitride-based heterostructure **14** includes a nitride-based semiconductor layer **140** and a nitride-based semiconductor layer **141**. The nitride-based semiconductor layer **141** is disposed on the substrate **13**, and the nitride-based semiconductor layer **140** is disposed on the nitride-based semiconductor layer **141**.
[0068] In this embodiment, a band gap of the nitride-based semiconductor layer **140** is different from a bandgap of the nitride-based semiconductor layer **141**, and a heterojunction **142** is formed in-between. In other words, an interface is formed between the nitride-based semiconductor layers **140**, **141**, and the heterojunction **142** is formed. The interface between the nitride-based semiconductor layers **140**, **141** is parallel with the interface between the substrates **12**, **13**.
[0069] For example, the nitride-based semiconductor layer **140** may comprise AlGaN, and the nitride-based semiconductor layer **141** may comprise GaN. The bandgap of the nitride-based semiconductor layer **140** is greater than the bandgap of the nitride-based semiconductor layer **141**. Therefore, a 2DEG region **143** is formed.
[0070] In one aspect, the nitride-based semiconductor circuit **1**A comprises an interlayer dielectric (ILD) layer **17** and an intermetal dielectric (IMD) layer **18**. The ILD layer **17** is disposed above the nitride-based heterostructure **14**, and the IMD layer **18** is disposed on the ILD layer **17**. To be specific, the ILD layer **17** is disposed on the passivation layers **16** and the connectors **15**.
[0071] In this embodiment, both the interconnections **190**, **191** pass through the ILD layer **17**, and the ILD layer **17** covers part of the connectors **15** and the nitride layer **161**. For example, the ILD layer **17** may include silicon oxide.
[0072] Also, the interconnection **190** is covered by the IMD layer **18** with the patterned conductive layer **20**, and only the interconnection **191** passes through the IMD layer **18**. For example, the IMD layer **18** may include silicon oxide.
[0073] In one aspect, the nitride-based semiconductor circuit **1**A comprises a plurality of vias **22**

and a plurality of vias **23**. The vias **22** pass through the ILD layer **17**, and the vias **22** connect the connectors **15** and the patterned conductive layer **20**.
[0074] The vias **23** passes through the IMD layer **18**, and the vias **23** connect the patterned conductive layer **20** and the patterned conductive layer **21**.
[0075] The vias **22**, **23** may include tungsten, but the disclosure is not limited thereto. The vias **22**, **23** can include, for example but are not limited to, metals, alloys, doped semiconductor materials (such as doped crystalline silicon), compounds such as silicides and nitrides, other conductor materials, or combinations thereof. The exemplary materials of the vias can include, for example but are not limited to, Ti, AlSi, TiN, or combination thereof. The vias may be a single layer, or plural layers of the same or different composition. In some embodiments, the vias form ohmic contacts with connectors **15** or the patterned conductive layers **20**, **21**. The ohmic contacts can be achieved by applying Ti, Al, or other suitable materials to the vias **22**, **23**. In some embodiments, each of the vias **22**, **23** is formed by at least one conformal layer and a conductive filling. The conformal layer can wrap the conductive filling. The exemplary materials of the conformal layer, for example but are not limited to, Ti, Ta, TiN, Al, Au, AlSi, Ni, Pt, or combinations thereof. The exemplary materials of the conductive filling can include, for example but are not limited to, AlSi, AlCu, or combinations thereof.
[0076] In one aspect, the nitride-based semiconductor circuit **1**A comprises a passivation layer **24**, and the passivation layer **24** covers part of the patterned conductive layer **21**.
[0077] The exemplary materials of the passivation layer **24** can include, for example but are not limited to, SiN.sub.x, SiO.sub.x, SiON, SiC, SiBN, SiCBN, oxides, nitrides, or combinations thereof. In some embodiments, the passivation layer **24** is a multi-layered structure, such as a composite dielectric layer of Al.sub.2O.sub.3/SiN, Al.sub.2O.sub.3/SiO.sub.2, AlN/SiN, AlN/SiO.sub.2, or combinations thereof.
[0078] FIGS. **3-16** are cross-sectional views of steps of manufacturing method of the nitride-based semiconductor circuit **1**A. Referring to FIG. **3**, the manufacturing method of the nitride-based semiconductor circuit **1**A includes: providing a semiconductor substrate **12** having dopants. To be specific, the substrate **12** is disposed on the optional semiconductor substrate **11**, but the disclosure is not limited thereto. In some embodiment, the substrate **12** may be provided without disposing on the optional substrate **11**.
[0079] Referring to FIG. **5**, the manufacturing method may include: disposing the semiconductor substrate **13** having dopants different from the dopants of the semiconductor substrate **12**. The pn junction **121** is formed between the substrates **12**, **13**.
[0080] Moreover, referring to FIG. **4**, in this embodiment, before the step of disposing the substrate **13**, a heavily doped area **120** is formed through doping selected a type of ion. For example, if the substrate **12** has n-type dopants, the heavily doped area **120** is formed through doping n-type ions for example, by ion implantation.
[0081] In some embodiments, the substrate **13** may be disposed without forming a heavily doped area **120**.
[0082] Moreover, referring to FIG. **5**, the substrates **11**, **13** may include p-type dopants, while the substrate **12** has n-type dopants, and the substrates **11-13** form a p-n-p structure (it is understood that, alternatively, the substrates **11** and **13** may include n-type dopants and the substrate **12** includes p-type dopants to form an n-p-n structure).
[0083] Referring to FIG. **6**, after disposing the substrate **13**, the manufacturing method includes: disposing a nitride-based heterostructure **14** on the semiconductor substrate **13**.
[0084] FIG. **7-11** depict the formation of the source, gate, and drain electrodes and associated conformal layers. These FIGS. will be discussed in further detail below.
[0085] Referring to FIG. **12**, after disposing the nitride-based heterostructure **14**, the manufacturing method include: disposing the connectors **15** on the nitride-based heterostructure **14**. The connectors **15** are disposed in two adjacent areas.

[0086] To be specific, the drain connector **150**, the gate connector **152**, and the source connector **151** are disposed on the area for HEMT structure **1***a*, and the drain connector **154**, the gate connector **155**, and the source connector **153** are disposed on the area of HEMT structure **1***b*. The areas of the HEMT structures **1***a*, **1***b* are adjacent.
[0087] Referring to FIG. **13**, after disposing the connectors **15**, the manufacturing method includes: etching through part of the nitride-based heterostructure **14**. Part of the nitride-based heterostructure **14** that is not covered by the connectors **15** is etched. The etched opening will expose part of the substrate **12**. To be specific, the heavily doped area **120** is exposed by the opening.
[0088] Referring to FIG. **14**, after etching the nitride-based heterostructure **14**, the manufacturing method includes: disposing the interconnection **190**; and disposing the patterned conductive layer **20**. The interconnection **190** is disposed in the etched openings that pass through the nitride-based heterostructure **14**. Part of the patterned conductive layer **20** is connected to the interconnection **190**, and the interconnection **190** electrically connects the substrate **12** to one of the connectors **15** through the patterned conductive layer **20**. For example, the interconnection **190** electrically connects the substrate **12** to the drain connector **150**.
[0089] Referring to FIG. **15**, after disposing the interconnection **190** and the patterned conductive layer **20**, the manufacturing method includes: etching another part of the nitride-based heterostructure **14**. Another part of the nitride-based heterostructure **14** that is not covered by the connectors **15** or filled with interconnection **190** is etched. The etched opening will expose part of the substrate **13**.
[0090] Referring to FIG. **16**, after etching the nitride-based heterostructure **14** for the second time, the manufacturing method includes: disposing the interconnection **191**; and disposing the patterned conductive layer **21** on the nitride-based heterostructure **14**. The interconnection **191** is disposed in the etched openings that pass through the nitride-based heterostructure **14** and expose the substrate **13**. Part of the patterned conductive layer **21** is connected to the interconnection **191**, and the interconnection **191** electrically connects the substrate **13** to another one of the connectors **15** through the patterned conductive layer **21**. For example, the interconnection **191** electrically connects the substrate **13** to the source connectors **151**, **153**.
[0091] The manufacturing method forms the pn junction **121** between the substrates **12**, **13**, and the interconnections **190**, **191** are electrically connected to the substrate **12**, **13** respectively. The pn junction **121** can be reverse biased or zero biased through connectors **15** that are electrically connected to the substrates **12**, **13**, and substrate voltage of different areas can be isolated.
[0092] In some embodiments, the interconnections **190**, **191** are electrically connected to external power sources, and the pn junction **121** can be reverse biased or zero biased through the external power sources.
[0093] Referring to FIG. **3**, the substrate **12** may be disposed on the substrate **11** through epitaxial growth techniques, while the dopants of the substrate **12** may be different from the dopants of the substrate **11**. In the embodiment, before providing the substrate **13** on the substrate **12**, the manufacturing method provides the substrate **11** and grows the substrate **12** on the substrate **11**, and the substrate **11** has the same dopant as the dopant of the substrate **13** as shown in FIG. **5**.
[0094] Referring to FIG. **4**, the heavily doped areas **120** may be formed through ion implantation. For example, n-type ions are implants in the heavily doped areas **120**. Moreover, the heavily doped areas **120** are formed through ion implantation and diffusion through high temperature. For example, the ion can include silicon, phosphorus, arsenic, and antimony. Also, by “heavily doped” it is meant a doping level of p-type or n-type dopants at a concentration of at least 10.sup.16 cm.sup.−3 or greater.
[0095] Referring to FIG. **5**, the substrate **13** may be provided on the substrate **12** through another epitaxial growth techniques. In this embodiment, the interface between the substrate **11** and the substrate **12** and the interface between the substrate **12** and the substrate **13** are parallel.

[0096] Referring to FIG. **6**, the nitride-based heterostructure **14** is disposed on the substrate **13**, and a doped nitride-based semiconductor layer **156** is formed on the nitride-based heterostructure **14**.
[0097] To be specific, the step of disposing the nitride-based heterostructure **14** on the substrate **13** includes: growing the nitride-based semiconductor layer **141** on the substrate **13**; and growing the nitride-based semiconductor layer **140** on the nitride-based semiconductor layer **141**. The bandgaps of the nitride-based semiconductor layers **140**, **141** are different, and the heterojunction **142** is formed with the 2DEG region **143**.
[0098] The doped nitride-based semiconductor layer **156** is formed through epitaxial growth techniques. In other words, the nitride-based semiconductor layer **141**, the nitride-based semiconductor layer **140**, and the doped nitride-based semiconductor layer **1522** are formed through epitaxial growth techniques respectively.
[0099] Referring to FIG. **7**, a conductive layer **157** is formed on the doped nitride-based semiconductor layer **156** through deposition. The conductive layer **157** can include, for example but is not limited to, metals, alloys, doped semiconductor materials (such as doped crystalline silicon), compounds such as silicides and nitrides, other conductor materials, or combinations thereof. The exemplary materials of the conductive layer **157** can include, for example but are not limited to, Ti, AlSi, TiN, or combinations thereof.
[0100] In this manufacturing method of the nitride-based semiconductor circuit **1**A, the step of disposing the connectors **15** comprises: disposing the gate connector **152** and the gate connector **155** on the nitride-based heterostructure **14**. Referring to FIG. **8**, the conductive layer **157** and the doped nitride-based semiconductor layer **156** are etched, and the gate connectors **152**, **155** are formed. The gate connector **152** includes the gate electrode **1521** and doped nitride-based semiconductor layer **1522**, and the gate connector **155** includes the gate electrode **1551** and doped nitride-based semiconductor layer **1552**. In other words, the manufacturing method etches the conductive layer **157** and the doped nitride-based semiconductor layer **156** into the gate connectors **152**, **155**.
[0101] Referring to FIG. **9**, the oxide layer **160** is disposed on the gate connector **155**. To be specific, the oxide layer **160** covers the top surface of the gate electrode **1551** of the gate connector **155**, and the oxide layer **160** covers side surfaces of the gate electrode **1551** and side surfaces of the doped nitride-based semiconductor layer **1552**. Furthermore, the oxide layer **160** covers a part of the nitride-based semiconductor layer **140** that is adjacent to the gate connector **155**.
[0102] In this manufacturing method of the nitride-based semiconductor circuit **1**A, after disposing the gate connectors **152**, **155**, the step of disposing the connectors **15** comprises: disposing the oxide layer on the gate connectors **152**, **155** and nitride-based heterostructure **14**; and etching a part of the oxide layer in contact with the gate connector **152**. In this embodiment, the step of disposing the oxide layer **160** may include: disposing a layer of oxide, and the layer covers the nitride-based heterostructure **14** and the gate connectors **152**, **155**; defining a pattern covering the gate connector **155** with photoresist; etching the rest of the layer of oxide which is not covered by the photoresist; and removing the photoresist on the gate connector **155**.
[0103] In this manufacturing method of the nitride-based semiconductor circuit **1**A, after etching the oxide layer, the step of disposing the connectors **15** comprises: disposing the nitride layer **161** on the gate connector **152** and the gate connector **155** and the oxide layer **160**. Referring to FIG. **10**, the nitride layer **161** is disposed on the gate connectors **152**, **155** and the nitride-based semiconductor layer **140**. The nitride layer **161** encapsulates the oxide layer **160** on the gate connector **155**, and the oxide layer **160** only touches the gate electrode **1551**, the doped nitride-based semiconductor layer **1552**, and the nitride layer **161** directly. The gate connector **152** is free from the oxide layer **160**.
[0104] In this embodiment, the oxide layer **160** elevates the height of the nitride layer **161** on the gate connector **155**, and the top surface of the nitride layer **161** on the gate connector **155** is higher than the top surface of the nitride layer **161** on the gate connector **152**. On the gate connector **155**,

the oxide layer **160** and the nitride layer **161** are overlapped, and the concentrations of hydrogen in the oxide layer **160** and the nitride layer **161** are different.

[0105] Referring to FIG. **11**, the manufacturing method etches the nitride layer **161**. A plurality of openings **162** are formed, and the openings **162** expose the nitride-based semiconductor layer **140**. The etching pattern of the nitride layer **161** can be defined by photoresist.

[0106] Referring to FIG. **12**, the manufacturing method forms the drain connector **150**, the source connector **151**, the drain connector **154**, and the source connector **153**. The drain connectors **150**, **154**, and the source connectors **151**, **153** include conductive metal and form ohmic contacts to the nitride-based semiconductor layer **140**, and the HEMT structure **1***a* and the HEMT structure **1***b* are formed.

[0107] The gate connector **152** of the HEMT structure **1***a* is free from the oxide layer **160**, and the gate connector **155** of the HEMT structure **1***b* directly contacts the oxide layer **160**. Since the concentrations of hydrogen in the oxide layer **160** and the nitride layer **161** are different, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*. Therefore, the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally opening by a voltage spike or voltage leakage.

[0108] Referring to FIG. **13**, the ILD layer **17** is disposed on the connectors **15** and the nitride layer **161**, and the ILD layer is etched. To be specific, at first, the ILD layer **17**, the nitride layer **161**, the nitride-based heterostructure **14**, the substrate **13** and part of the substrate **12** are etched, and the openings **170** are formed. The openings **170** expose the substrate **12**, and the openings **170** expose the heavily doped area **120** of the substrate **12**. The rest of the ILD layer **17**, the nitride layer, and the oxide layer are etched, and the openings **171**, **172**, **173** are formed. The openings **171** pass through the ILD layer **17**, and the openings **171** expose the drain connectors **150**. **154**. The openings **172** pass through the ILD layer **17** and the passivation layers **16**, and the openings **172** expose the gate connectors **152**, **155**. The openings **173** pass through the ILD layer **17**, and the openings **173** expose the source connector **151**, **153**. Moreover, the manufacturing method disposed dielectric layers **192** on the side wall of the openings **170** before etching the openings **171**, **172**, **173**. For example, the dielectric layers **192** can include oxide or silicon nitride.

[0109] Referring to FIG. **14**, the manufacturing method deposits conductive material in the openings **170**, **171**, **172**, **173** as shown in FIG. **13**, and the interconnection **190** and the vias **22** are formed, and the interconnections **190** connect the heavily doped areas **120**. For example, the conductive material can include tungsten, but the disclosure is not limited thereto.

[0110] After the interconnection **190** and the vias **22** are formed, the patterned conductive layer **20** is disposed on the ILD layer **17**, and the patterned conductive layer **20** electrically connects the vias **22** and the interconnection **190**. In one aspect, the drain connector **150** is electrically connected to the substrate **12** through the conductive material of the interconnection **190** and the patterned conductive layer **20**.

[0111] Referring to FIG. **15**, the manufacturing method deposits the IMD layer **18** on the ILD layer **17** and the patterned conductive layer **20**. After disposing the IMD layer **18**, the IMD layer **18** is etched for the first time, and the etched openings expose parts of the patterned conductive layer **20**. The manufacturing method disposes conductive material in the openings, and the vias **23** are formed.

[0112] After the vias **23** are formed, the manufacturing method involves etching the IMD layer **18** again. The IMD layer **18**, the ILD layer **17**, the nitride layer **161**, the nitride-based heterostructure **14**, and part of the substrate **13** are etched, and openings **180** are formed. The openings **180** pass through the nitride-based heterostructure **14** and expose the substrate **13**.

[0113] Referring to FIG. **16**, the manufacturing method deposits the conductive material in the openings **180** as shown in FIG. **15**, and the interconnections **191** are formed. Moreover, the patterned conductive layer **21** is disposed on the IMD layer **18**, and the patterned conductive layer

21 electrically connects the vias **23** and the interconnection **191**.

[0114] In one aspect, the source connector **151** and the drain connector **154** are electrically connected to an area of the substrate **13**, which is surrounded by the interconnection **190**, through one of the interconnections **191**, and the source connector **153** is electrically connected to another area of the substrate **13** through another one of the interconnections **191**. Also, the patterned conductive layer **21** connects the HEMT structure **1***a* to the HEMT structure **1***b*, and the HEMT structures **1***a*, **1***b* can form a half bridge circuit.

[0115] After disposing the patterned conductive layer **21**, the manufacturing method creates the passivation layer **24** as shown in FIG. **2**. The passivation layer **24** include dielectric material, and the passivation layer **24** exposes only parts of the patterned conductive layer **21** for external connection.

[0116] FIG. **17** is a top view of a nitride-based semiconductor circuit according to another embodiment of the present disclosure, and FIG. **18** is a cross-sectional view of a semiconductor circuit according to cutting plane line I**2** in FIG. **17**, and parts of the layers of the semiconductor circuit are omitted in the top view for clarity. To be specific, FIG. **17** only shows a top view of a substrate **12**, a heavily doped area **120**, and part of a patterned conductive layer **21** thereon for clarity. The nitride-based semiconductor circuit **1**B is similar to the nitride-based semiconductor circuit **1**A, and the nitride-based semiconductor circuit **1**B has a nitride-based heterostructure **14** and connectors **15**, and the connectors **15** are disposed on the nitride-based heterostructure **14**, and the detailed description about the similar components will not be repeated.

[0117] In this embodiment, the nitride-based semiconductor circuit **1**B includes a substrate **13** and the substrate **12**, and the substrate **13** is disposed on the substrate **12**, and a pn junction **121** is formed. For example, the substrate **12** has p-type dopants, and the substrate **13** has n-type dopants, and the pn junction **121** is formed between the substrates **12**, **13**. The heavily doped area **120** is formed through doping a type of ion. For example, the substrate **12** has p-type dopants, and the heavily doped area **120** is formed through doping p-type ions. For example, the p-type ions include magnesium.

[0118] The nitride-based heterostructure **14** is disposed on the substrate **13**. The nitride-based heterostructure **14** has a nitride-based semiconductor layer **140**, and a nitride-based semiconductor layer **141**, and the nitride-based semiconductor layer **140** is disposed on the nitride-based semiconductor layer **141**, and a heterojunction **142** is formed with a 2DEG region **143**.

[0119] The connectors **15** are disposed on the nitride-based heterostructure **14**, and the connectors **15** has drain connectors **150**, **154**, gate connectors **152**, **155**, source connectors **151**, **153**. The drain connector **150**, the gate connector **152**, and the source connector **151** are disposed on the region of a HEMT structure **1***a*, and the drain connector **154**, the gate connector **155**, and the source connector **153** are disposed on the region of a HEMT structure **1***b*.

[0120] The nitride-based semiconductor circuit **1**B has a plurality of connecting vias **19**, and the connecting vias **19** include interconnection **190** and interconnections **191**. The interconnection **190** electrically connects the substrate **12** to the source connector **153**, and one of the interconnections **191** electrically connects the substrate **13** to the source connector **151** and the drain connector **154**, and another one of the interconnections **191** electrically connects the substrate **13** to the source connector **153**. Moreover, the interconnection **190** of this embodiment is electrically connected to the heavily doped area **120** of the substrate **12**. Therefore, the pn junction **121** can be reversely biased or zero biased through the connecting vias **19**, and the substrate voltage of the HEMT structure **1***a* can't affect the substrate voltage of the HEMT structure **1***b*.

[0121] In this embodiment, the heavily doped area **120** surrounds a projection **1***d* of the HEMT structure **1***b* on the substrate **12**, and a projection **1***c* of the HEMT structure **1***a* on the substrate **12** is not surrounded by the heavily doped area **120**. Therefore, the substrate voltage of the HEMT structure **1***a* is isolated from the substrate voltage of the HEMT structure **1***b*. In other words, the substrate **12** and the substrate **13** form a plurality of diode structures, and the diode structures surrounded by the heavily doped area **120** is isolated from the diode structures that is not surrounded by the heavily doped area **120**.

[0122] For example, when the HEMT structure **1***a* is on and the HEMT structure **1***b* is off, the pn junction **121** between the substrates **12**,**13** under the HEMT structure **1***a* is reverse biased, and the pn junction **121** between the substrates **12**,**13** under the HEMT structure **1***b* is zero biased, and a depletion region is created in the substrates **12**, **13** of the HEMT structure **1***a*. The voltage of the substrate **13** under the HEMT structure **1***b* is isolated while the voltage of the substrate **13** under the HEMT structure **1***a* is changing.

[0123] When the HEMT structure **1***a* is off and the HEMT structure **1***b* is on, the pn junction **121** between the substrates **12**, **13** under the HEMT structure **1***a* is reversely biased, and the pn junction **121** between the substrates **12**, **13** under the HEMT structure **1***b* is zero biased, and a depletion region is created in the substrates **12**, **13** of the HEMT structure **1***a*. The voltage of the substrate **13** under the HEMT structure **1***a* is isolated while the voltage of the substrate **12** under the HEMT structure **1***b* is changing.

[0124] In other words, the substrate **13** shares the same voltage with the source connector **151** or the source connector **153**, and the substrate **12** shares the same voltage with the source connector **153**, and voltages of the substrates **12**, **13** reversibly bias part of the pn junction **121** therebetween.

[0125] Also, the nitride-based semiconductor circuit **1**B includes passivation layers **16** covering the gate connectors **152**, **155**, while only the gate connector **155** is directly in touch with the oxide layer **160**. The gate connector **152** is only covered by the nitride layer **161**. Therefore, the threshold voltages of the HEMT structures **1***a*, **1***b* are different. The HEMT structure **1***a* can prevent from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally opening by a voltage spike or voltage leakage.

[0126] FIG. **19** is a top view of a nitride-based semiconductor circuit according to another embodiment of the present disclosure, and FIG. **20** is a cross-sectional view of the nitride-based semiconductor circuit according to cutting plane line I**3** in FIG. **19**, and parts of the layers of the semiconductor circuit are omitted in the top view for clarity. To be specific, FIG. **19** only shows a top view of a substrate **12**, a heavily doped area **120**, and part of a patterned conductive layer **21** thereon for clarity. The nitride-based semiconductor circuit **1**C is similar to the nitride-based semiconductor circuit **1**A, and the nitride-based semiconductor circuit **1**C has substrates **11-13**, a nitride-based heterostructure **14**, and connectors **15**, and the detailed description about the similar components will not be repeated. The nitride-based heterostructure **14** is disposed on the substrate **13**, and the connectors **15** are disposed on the nitride-based heterostructure **14**.

[0127] In this embodiment, the substrate **12** is provided on the substrate **11**, and the substrate **13** is provided on the substrate **12**, and a pn junction **121** is formed. For example, the substrate **11** has p-type dopants, and the substrate **12** has n-type dopants, and the substrate **13** has p-type dopants. The substrates **11-13** form a p-n-p structure, and the pn junction **121** is located between the substrate **12** and the substrate **13**, and the heavily doped area **120** is formed on the substrate **12**.

[0128] The nitride-based heterostructure **14** is disposed on the substrate **13**. The nitride-based heterostructure **14** has a nitride-based semiconductor layer **140**, and a nitride-based semiconductor layer **141**, and the nitride-based semiconductor layer **140** is disposed on the nitride-based semiconductor layer **141**, and a heterojunction **142** is formed with a 2DEG region **143**.

[0129] The connectors **15** are disposed on the nitride-based heterostructure **14**, and the connectors **15** has drain connectors **150**, **154**, gate connectors **152**, **155**, source connectors **151**, **153**. The drain connector **150**, the gate connector **152**, and the source connector **151** are disposed on the region of a HEMT structure **1***a*, and the drain connector **154**, the gate connector **155**, and the source connector **153** are disposed on the region of a HEMT structure **1***b*.

[0130] The nitride-based semiconductor circuit **1**C has a plurality of connecting vias **19**, and the connecting vias **19** include interconnection **190** and interconnection **191**. The interconnection **190** electrically connects the substrate **12** to the drain connector **150**, and one of the interconnections

191 electrically connects the substrate **13** to the source connector **151** and the drain connector **154**, and another one of the interconnections **191** electrically connects the substrate **13** to the source connector **153**. Moreover, the interconnection **190** of this embodiment is electrically connected to the heavily doped area **120** of the substrate **12**. Therefore, the pn junction **121** can be reversely biased or zero biased through the connecting vias **19**, and the substrate voltage of the HEMT structure **1***a* does not affect the substrate voltage of the HEMT structure **1***b*.

[0131] In this embodiment, the heavily doped area **120** surrounds both projections **1***c*, **1***d* of the HEMT structures **1***a*, **1***b* on the substrate **12**. Therefore, the substrate voltage of the HEMT structure **1***a* is isolated from the substrate voltage of the HEMT structure **1***b*. In other words, the substrate **12** and the substrate **13** form a plurality of diode structures, and all the diode structures under the HEMT structures **1***a*, **1***b* are surrounded by the heavily doped area **120**, and the diode structures are isolated into two groups.

[0132] For example, when the HEMT structure **1***a* is on and the HEMT structure **1***b* is off, the pn junction **121** between the substrates **12**, **13** under the HEMT structure **1***a* is zero biased, and the pn junction **121** between the substrates **12**, **13** under the HEMT structure **1***b* is reverse biased, and a depletion region is created in the substrates **12**, **13** of the HEMT structure **1***b*. The voltage of the substrate **13** under the HEMT structure **1***b* is isolated while the voltage of the substrate **13** under the HEMT structure **1***a* is changing.

[0133] When the HEMT structure **1***a* is off and the HEMT structure **1***b* is on, the pn junction **121** between the substrates **12**,**13** under the HEMT structures **1***a*, **1***b* are both reversely biased, and depletion regions are created in the substrates **12**, **13** of the HEMT structures **1***a*, **1***b*. The voltage of the substrate **13** under the HEMT structure **1***a* is isolated while the voltage of the substrate **13** under the HEMT structure **1***b* is changing.

[0134] In other words, the substrate **12** shares the same voltage with the drain connector **150**, and the substrate **13** shares the same voltage with the source connector **151** or the source connector **153**, and voltages of the substrates **12**, **13** reversibly bias the pn junction **121** therebetween.

[0135] Also, the nitride-based semiconductor circuit **1**C includes passivation layers **16** covering the gate connectors **152**, **155**, while only the gate connector **155** is directly in contact with the oxide layer **160**. The gate connector **152** is only covered by the nitride layer **161**. Therefore, the threshold voltages of the HEMT structures **1***a*, **1***b* are different. The HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally opening by a voltage spike or voltage leakage.

[0136] FIG. **21** is a cross-sectional view of a nitride-based semiconductor circuit according to another embodiment of the present disclosure. The nitride-based semiconductor circuit **1**D is similar to the nitride-based semiconductor circuit **1**C, and the detailed description regarding the similar components will not be repeated. The differences between the two circuits are: the nitride-based semiconductor circuit **1**C has only a substrate **12** and a substrate **13** being disposed on the substrate **12**. Also, the interconnection **190** electrically connects the heavily doped area **120** of the substrate **12** to the source connector **153**, and the interconnection **191** electrically connects the substrate **13** to the source connector **151** and the drain connector **154**.

[0137] For example, the substrate **12** has p-type dopants, and the substrate **13** has n-type dopants, and the source connector **153** is grounded. Therefore, the pn junction **121** under the HEMT structure **1***a* is always reverse biased, and the substrate voltages of the HEMT structures **1***a*, **1***b* are individual isolated.

[0138] FIG. **22** is a cross-sectional view of a nitride-based semiconductor circuit according to an embodiment of the present disclosure. The nitride-based semiconductor circuit **1**E has a nitride-based heterostructure **14**, connectors **15**, passivation layers **16**, and patterned conductive layers **20**, **21**, which are similar to the nitride-based semiconductor circuit **1**A, and the detailed description regarding the similar components will not be repeated.

[0139] In this embodiment, the nitride-based semiconductor circuit **1**E has substrates **11-13**, and the

substrate **12** is disposed on the substrate **11**, and the substrate **13** is disposed on the substrate **12**. A heavily doped area **120** is formed, and the heavily doped area **120** is formed across the substrates **12**, **13**. The heavily doped area **120** is extended from the top surface of the substrate **13** to the inside of the substrate **12**. In other words, part of the substrate **12** and part of the substrate **13** thereon form the heavily doped area **120** together, and the heavily doped area **120** is formed after the substrate **13** is disposed on the substrate **12**.
[0140] For example, the substrate **13** has p-type dopants, and the substrate **12** has n-type dopants, and the heavily doped area **120** is formed through implantation of n-type ions. Therefore, a pn junction is formed between the substrate **13** and the heavily doped area **120**, and no current will be transmitted between the heavily doped area **120** and the substrate **13**.
[0141] The interconnection **190** is electrically connected to the heavily doped area **120**, and the interconnection **190** in the substrate **13** is surrounded by the pn junction between the substrate **13** and the heavily doped area **120**, and the current from the interconnection **190** can be transmitted to the substrate **12** without leakage in the substrate **13**.
[0142] A dielectric layer **192** is disposed on the side wall of the opening where the interconnection **190** is disposed. In some embodiment, the interconnection **190** can be directly disposed in the nitride-based semiconductor circuit **1**E without being surrounded by the dielectric layer **192**.
[0143] FIG. **23** is a cross-sectional view of a nitride-based semiconductor circuit according to an embodiment of the present disclosure. The nitride-based semiconductor circuit **1**F has a nitride-based heterostructure **14**, connectors **15**, passivation layers **16**, and patterned conductive layers **20**, **21**, which are similar to the nitride-based semiconductor circuit **1**B, and the detailed description regarding the similar components will not be repeated.
[0144] In this embodiment, the nitride-based semiconductor circuit **1**F has substrates **12**,**13**, and the substrate **13** is disposed on the substrate **12**. A heavily doped area **120** is formed, and the heavily doped area **120** is formed across the substrates **12**, **13**. The heavily doped area **120** is extended from the top surface of the substrate **13** to the inside of the substrate **12**. In other words, part of the substrate **12** and part of the substrate **13** thereon form the heavily doped area **120** together, and the heavily doped area **120** is formed after the substrate **13** is disposed on the substrate **12**.
[0145] For example, the substrate **13** has n-type dopants, and the substrate **12** has p-type dopants, and the heavily doped area **120** is formed through implantation of p-type ions. Therefore, pn junction is formed between the substrate **13** and the heavily doped area **120**, and no current will be transmitted between the heavily doped area **120** and the substrate **13**.
[0146] The interconnection **190** is electrically connected to the heavily doped area **120**, and the interconnection **190** in the substrate **13** is surrounded by the pn junction between the substrate **13** and the heavily doped area **120**, and the current from the interconnection **190** can be transmitted to the substrate **12** without leakage in the substrate **13**.
[0147] A dielectric layer **192** is disposed on the side wall of the opening where the interconnection **190** is disposed. In some embodiments, the interconnection **190** can be directly disposed in the nitride-based semiconductor circuit **1**F without being surrounded by the dielectric layer **192**.
[0148] FIG. **24** is a cross-sectional view of a nitride-based semiconductor circuit according to an embodiment of the present disclosure. The nitride-based semiconductor circuit **1**G has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, and patterned conductive layers **20**, **21**, which are similar to the nitride-based semiconductor circuit **1**A, and the detailed description regarding the similar components will not be repeated.
[0149] In this embodiment, the nitride-based semiconductor circuit **1**G has passivation layers **16**, and the passivation layers **16** include an oxide layer **160** and a nitride layer **161**. The oxide layer **160** is in contact with the gate connector **155** and isolated from the gate connector **152**, and the nitride layer is in contact with the gate connector **152**.
[0150] To be specific, the gate connector **155** has a doped nitride-based semiconductor layer **1552** and a gate electrode **1551**, and the gate electrode **1551** is disposed on the doped nitride-based

semiconductor layer **1552**. The oxide layer **160** covers the side surface of the doped nitride-based semiconductor layer **1552**, and the doped nitride-based semiconductor layer **1552** makes contact with the oxide layer **160**. The top surface of the oxide layer **160** is not covered by the nitride layer **161**, and the nitride layer **161** only covers lower part of side surface of the oxide layer **160**. The ILD layer **17** makes contact with the oxide layer **160** and the nitride layer **161**. Therefore, the HEMT structures **1***a*, **1***b* may have different threshold voltages. The HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally opening by a voltage spike or voltage leakage.
[0151] FIGS. **25-28** are cross-sectional views of steps of manufacturing the nitride-based semiconductor circuit **1**G, and FIGS. **25-28** illustrate the manufacturing method of the passivation layers **16** of the nitride-based semiconductor circuit **1**G, and the remaining steps of the manufacturing method are substantially similar to the embodiment shown in FIGS. **3-16**, discussed in detail above.
[0152] Referring to FIG. **25**, the substrate **12** is disposed on the substrate **11**, and the substrate **12** has a heavily doped area **120**. The substrate **13** is disposed on the substrate **12**, and the pn junction **121** is formed between the substrates **12**, **13**.
[0153] The nitride-based heterostructure **14** is disposed on the substrate **13**, and the nitride-based heterostructure **14** has the nitride-based semiconductor layers **140**, **141**. A heterojunction **142** is formed between the nitride-based semiconductor layers **140**, **141** with the 2DEG region **143**.
[0154] The gate connectors **152**, **155** are disposed on the nitride-based heterostructure **14**, and the nitride layer **161** is disposed on the gate connectors **152**, **155** and the nitride-based heterostructure **14**. For example, the nitride layer **161** may comprise silicon nitride, and the nitride layer **161** may be formed through LPCVD. In this step, both the gate connectors **152**, **155** make contact with the nitride layer **161**.
[0155] Referring to FIG. **26**, the nitride layer **161** is etched. This step etches the nitride layer in contact with the gate connector **155**, and the side surface and the top surface of the gate connector **155** are exposed. Furthermore, openings **1610** are formed around the gate connector **155**, and the openings **1610** expose the top surface of the nitride-based semiconductor layer **140**. At this step, the nitride layer **161** that covered the surface of the doped nitride-based semiconductor layer **1552** is etched.
[0156] Referring to FIG. **27**, the oxide layer **160** is disposed on the nitride layer **161** and the gate connector **155**, and the oxide layer **160** fills the openings **1610** as shown in FIG. **26**. The oxide layer **160** is applied through PECVD, and the oxide layer **160** makes contact with the gate connector **155**.
[0157] Referring to FIG. **28**, the oxide layer **160** is etched, and only the oxide layer **160** that is in contact with the gate connector **155** is remained. The oxide layer **160** near the gate connector **152** is etched. In this embodiment, the gate connector **155** is only covered by the oxide layer **160**.
[0158] The hydrogen concentration of oxide layer **160** is different from the hydrogen concentration of nitride layer **161**, and the doped nitride-based semiconductor layer **1552** is in direct contact with the oxide layer **160**. While the doped nitride-based semiconductor layers **1522**, **1552** are doped with p-type dopant, activation of magnesium in the doped nitride-based semiconductor layer **1522** and activation of magnesium in the doped nitride-based semiconductor layer **1552** can be different during exposure to a high temperature thermal environment. Therefore, the threshold voltage of the HEMT structure **1***a* is different from the threshold voltage of the HEMT structure **1***b* as shown in FIG. **24**. The HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is open. In other words, the HEMT structures **1***a*, **1***b* can be prevented from accidentally opening due to a voltage spike or voltage leakage.
[0159] In other words, in some embodiments of the present disclosure, the oxide layer **160** can be formed before the deposition of the nitride layer **161**. In some other embodiments of the present disclosure, the oxide layer **160** can be formed after the deposition of the nitride layer **161**.

[0160] FIG. **29** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure. The nitride-based semiconductor circuit **1**H has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, patterned conductive layers **20**, **21**, and connecting vias **19**, which are similar to the nitride-based semiconductor circuit **1**A, and the detailed description regarding the similar components will not be repeated.

[0161] In this embodiment, part of the oxide layer **160** on the gate connector **155** is etched. To be specific, parts of the side surface and the top surface of the gate electrode **1551** of the gate connector **155** are free from coverage of the oxide layer **160**. Part of the side surface of the doped nitride-based semiconductor layer **1552** is free from coverage of the oxide layer **160**, and the rest of the side surface of the doped nitride-based semiconductor layer **1552** is in contact with the oxide layer **160**. Therefore, the oxide layer **160** may be reduced in size, and the threshold voltage of the HEMT structure **1***b* and the threshold voltage of the HEMT structure **1***a* are different. For example, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*, and the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is turned on.

[0162] FIG. **30** is a cross-sectional view of a nitride-based semiconductor circuit according to an embodiment of the present disclosure. The nitride-based semiconductor circuit **1**J has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, patterned conductive layers **20**, **21**, and connecting vias **19**, which are similar to the nitride-based semiconductor circuit **1**A, and the detailed description regarding the similar components will not be repeated.

[0163] In this embodiment, part of the oxide layer **160** on the gate connector **155** is etched. To be specific, the top surface and part of the side surface of the gate electrode **1551** of the gate connector **155** are free from coverage of the oxide layer **160**. Part of the side surface of the doped nitride-based semiconductor layer **1552** is free from coverage of the oxide layer **160**, and the rest of the side surface of the doped nitride-based semiconductor layer **1552** is in contact with the oxide layer **160**. Therefore, the oxide layer **160** may reduce its size, and the threshold voltage of the HEMT structure **1***b* and the threshold voltage of the HEMT structure **1***a* are different. For example, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*, and the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is turned on.

[0164] FIG. **31** is a cross-sectional view of a nitride-based semiconductor circuit according to an embodiment of the present disclosure. The nitride-based semiconductor circuit **1**K has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, patterned conductive layers **20**, **21**, and connecting vias **19**, which are similar to the nitride-based semiconductor circuit **1**A, and the detailed description regarding the similar components will not be repeated.

[0165] In this embodiment, part of the oxide layer **160** on the gate connector **155** is etched. To be specific, side surface and the top surface of the gate electrode **1551** of the gate connector **155** are free from coverage of the oxide layer **160**. Part of the side surface of the doped nitride-based semiconductor layer **1552** is free from coverage of the oxide layer **160**, and the rest of the side surface of the doped nitride-based semiconductor layer **1552** is in contact with the oxide layer **160**. Therefore, the oxide layer **160** may reduce its size, and the threshold voltage of the HEMT structure **1***b* and the threshold voltage of the HEMT structure **1***a* are different. For example, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*, and the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is turned on.

[0166] FIG. **32** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure. The nitride-based semiconductor circuit **1**L has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, patterned conductive layers **20**, **21**, and connecting vias **19**, which are similar to the nitride-based semiconductor circuit **1**G, and the detailed description regarding the similar components will not be repeated.

[0167] In this embodiment, the oxide layer **160** is disposed after the deposition and etching of the nitride layer **161**. Part of the oxide layer **160** on the gate connector **155** is etched. To be specific, parts of the top surface and side surface of the gate electrode **1551** of the gate connector **155** are free from coverage of the oxide layer **160**. Part of the side surface of the doped nitride-based semiconductor layer **1552** is free from coverage of the oxide layer **160**, and the rest of the side surface of the doped nitride-based semiconductor layer **1552** is in contact with the oxide layer **160**. Therefore, the oxide layer **160** may reduce its size, and the threshold voltage of the HEMT structure **1***b* and the threshold voltage of the HEMT structure **1***a* are different. For example, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*, and the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is turned on.
[0168] FIG. **33** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure. The nitride-based semiconductor circuit **1**M has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, patterned conductive layers **20**, **21**, and connecting vias **19**, which are similar to the nitride-based semiconductor circuit **1**G, and the detailed description regarding the similar components will not be repeated.
[0169] In this embodiment, the oxide layer **160** is disposed after the deposition and etching of the nitride layer **161**. Part of the oxide layer **160** on the gate connector **155** is etched. To be specific, the top surface and part of the side surface of the gate electrode **1551** of the gate connector **155** are free from coverage of the oxide layer **160**. Part of the side surface of the doped nitride-based semiconductor layer **1552** is free from coverage of the oxide layer **160**, and the rest of the side surface of the doped nitride-based semiconductor layer **1552** is in contact with the oxide layer **160**. Therefore, the oxide layer **160** may be reduced in size, and the threshold voltage of the HEMT structure **1***b* and the threshold voltage of the HEMT structure **1***a* are different. For example, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*, and the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is turned on.
[0170] FIG. **34** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure. The nitride-based semiconductor circuit **1**N has substrates **11-13**, a nitride-based heterostructure **14**, connectors **15**, patterned conductive layers **20**, **21**, and connecting vias **19**, which are similar to the nitride-based semiconductor circuit **1**G, and the detailed description regarding the similar components will not be repeated.
[0171] In this embodiment, the oxide layer **160** is disposed after the deposition and etching of the nitride layer **161**. Part of the oxide layer **160** on the gate connector **155** is etched. To be specific, the gate electrode **1551** is isolated from the oxide layer **160**, and the side surface and the top surface of the gate electrode **1551** of the gate connector **155** are free from coverage of the oxide layer **160**. Part of the side surface of the doped nitride-based semiconductor layer **1552** is free from coverage of the oxide layer **160**, and the rest of the side surface of the doped nitride-based semiconductor layer **1552** is in contact with the oxide layer **160**. Therefore, the oxide layer **160** may be reduced in size, and the threshold voltage of the HEMT structure **1***b* and the threshold voltage of the HEMT structure **1***a* are different. For example, the threshold voltage of the HEMT structure **1***b* is higher than the threshold voltage of the HEMT structure **1***a*, and the HEMT structure **1***a* can be prevented from opening while the HEMT structure **1***b* is turned on.
[0172] FIG. **35** is a top view of a nitride-based semiconductor circuit **3**A according to some embodiments of the present disclosure, and FIG. **36** is a cross-sectional view of the nitride-based semiconductor circuit **3**A according to cutting plane line I**4** in FIG. **35**, and parts of layers of the nitride-based semiconductor circuit **3**A are omitted in the top view for clarity. FIG. **35** only shows top view of a substrate structure **30**, and a patterned conductive layer **35**.
[0173] The nitride-based semiconductor circuit **3**A includes a substrate structure **30**, a nitride-based heterostructure **31**, a plurality of connectors **32**, and a plurality of connecting vias **38**. The substrate

structure **30** includes a semiconductor substrate **300**, and a semiconductor substrate **301**. The semiconductor substrate **300** has a region **3***a* and region **3***b*. The region **3***a* is adjacent to the region **3***b*. The semiconductor substrate **301** is embedded in the region **3***b* of the semiconductor substrate **300**. A top surface **3000** of the semiconductor substrate **300** and a top surface **3010** of the semiconductor substrate **301** are coplanar. The semiconductor substrate **300** has dopants, and the semiconductor substrate **301** has dopants that are different from the dopants of the semiconductor substrate **300**, and a pn junction **302** is formed between the semiconductor substrates **300**, **301**. For example, the semiconductor substrates **300**, **301** may include silicon.

[0174] The nitride-based heterostructure **31** is disposed on the substrate structure **30**, and the nitride-based heterostructure **31** is disposed on both the semiconductor substrate **300** and the semiconductor substrate **301**. The connectors **32** are disposed on the nitride-based heterostructure **31**. The connecting vias **38** go through the nitride-based heterostructure **31**. The connecting vias **38** includes interconnection **380** and interconnection **381**. The interconnection **380** electrically connects the region **3***a* of the semiconductor substrate **300** to one of the connectors **32**, and the interconnection **381** electrically connects the semiconductor substrate **301** to another one of the connectors **32**.

[0175] In some embodiments, the connectors **32** and connecting vias **38** can include, for example but are not limited to, metals, alloys, doped semiconductor materials (such as doped crystalline silicon), compounds such as silicides and nitrides, other conductor materials, or combinations thereof. The exemplary materials of the connectors **32** and connecting vias **38** can include, for example but are not limited to, W, Au, Pd, Ta, Co, Ni, Pt, Mo, Ti, AlSi, TiN, or combination thereof. The connectors **32** and connecting vias **38** may be a single layer, or plural layers of the same or different composition. In some embodiments, the connectors **32** and connecting vias **38** form ohmic contacts with the nitride-based heterostructure **31** or semiconductor substrates **300**, **301**. The ohmic contacts can be achieved by applying Ti, Al, or other suitable materials to the connectors **32** and connecting vias **38**. In some embodiments, each of the connectors **32** and connecting vias **38** is formed by at least one conformal layer and a conductive filling. The conformal layer can surround the conductive filling. The exemplary materials of the conformal layer, for example but are not limited to, Ti, Ta, TiN, Al, Au, AlSi, Ni, Pt, or combinations thereof. The exemplary materials of the conductive filling can include, for example but are not limited to, AlSi, AlCu, or combinations thereof.

[0176] In this embodiment, the pn junction **302** separate the substrate structure **30** into two parts, and some of the connectors **32** are disposed above the semiconductor substrate **301**, and some of the connectors **32** are disposed above the region **3***a* of the semiconductor substrate **300**. The substrate voltage of the semiconductor substrate **301** does not affect the substrate voltage of the semiconductor substrate **300**. Moreover, the interconnection **381** is electrically connected to the semiconductor substrate **301**, and the interconnection **380** is electrically connected to the semiconductor substrate **300**. Therefore, the connecting vias **38** can reversely bias the pn junction **302**, and a depletion region is formed, and voltage of the semiconductor substrate **300** can be isolated from voltage of the semiconductor substrate **301**.

[0177] The semiconductor substrate **301** is embedded in the semiconductor substrate **300**, and an interface between the semiconductor substrate **300** and the semiconductor substrate **301** is concave or concave-like from a cross-sectional perspective. Moreover, the pn junction **302** is formed along the interface, and the interface is concave or concave-like. Therefore, the semiconductor substrate **301** can provide a sufficient area and thickness for deposition of some of the connectors **32**.

[0178] To be specific, the connectors **32** includes drain connectors **320**, **324**, gate connectors **322**, **325**, and source connectors **321**, **323**. The gate connector **322** is located between the drain connector **320** and the source connector **321**, and the gate connector **325** is located between the drain connector **324** and the source connector **323**.

[0179] The nitride-based heterostructure **31** includes a nitride-based semiconductor layer **310** and a nitride-based semiconductor layer **311**. The nitride-based semiconductor layer **310** is disposed on the nitride-based semiconductor layer **311**, and the material of the nitride-based semiconductor layer **310** is different from the material of the nitride-based semiconductor layer **311**, and a heterojunction **312** is formed. Moreover, the band gap of the nitride-based semiconductor layer **310** is different from the band gap of the nitride-based semiconductor layer **311**, and a 2DEG region **313** is formed.

[0180] The drain connector **320**, the gate connector **322**, and the source connector **321** are located above the region **3***b* of the semiconductor substrate **300**, and the connectors **320-322** form a HEMT structure with part of the nitride-based heterostructure **31**. To be specific, the drain connector **320**, the gate connector **322**, and the source connector **321** are disposed on the semiconductor substrate **301**.

[0181] The drain connector **324**, the gate connector **325**, and the source connector **323** are located above the region **3***a* of the semiconductor substrate **300**, and the connectors **323-325** form a HEMT structure with another part of the nitride-based heterostructure **31**.

[0182] The pn junction **302** is formed between the semiconductor substrate **301** and the semiconductor substrate **300**, and the pn junction **302** can be reversely biased or zero biased. Therefore, substrate voltage of the HEMT structure above the region **3***b* can be isolated from substrate voltage of the HEMT structure above the region **3***a*.

[0183] In one aspect, the nitride-based semiconductor circuit **3**A includes a patterned conductive layer **35**. The patterned conductive layer **35** is disposed on the connectors **32**, and the patterned conductive layer **35** electrically connects the connectors **32**. The patterned conductive layer **35** has a plurality of pads, and the pad **350** electrically connects the source connector **321** and the drain connector **324**. In other words, the patterned conductive layer **35** electrically connects the HEMT structure above the region **3***b* to the HEMT structure above the region **3***a*.

[0184] In an embodiment, the nitride-based semiconductor circuit **3**A can form a half bridge circuit, and the HEMT structure above the region **3***b* is connected at the high side, and the HEMT structure above the region **3***a* is connected at the low side.

[0185] Furthermore, the pad **350** electrically connected to the source connector **321** and the drain connector **324** overlays part of the pn junction **302** and electrically connects to the semiconductor substrate **301**. Therefore, the patterned conductive layer **35** electrically connects the HEMT structures, and the HEMT structures of the nitride-based semiconductor circuit **3**A can be electrically connected while the substrate voltages of the HEMT structures are separate.

[0186] In one aspect, the pad **350** electrically connected to the source connector **321** and the drain connector **324** overlays both the region **3***a* and the region **3***b*. Therefore, the HEMT structures of the nitride-based semiconductor circuit **3**A can be electrically connected while the substrate voltages of the HEMT structures are separate.

[0187] In one aspect, the nitride-based semiconductor circuit **3**A includes a patterned conductive layer **34**. The patterned conductive layer **34** is disposed between the connectors **32** and the patterned conductive layer **35**. The patterned conductive layer **34** electrically connects the connectors **32** to the patterned conductive layer **35**. Therefore, the patterned conductive layer **34** may provide proper connection between the patterned conductive layer **35** and the connectors **32**.

[0188] In this embodiment, the interconnection **380** electrically connects the semiconductor substrate **300** to the source connector **323**, and the interconnection **381** electrically connects the semiconductor substrate **301** to the source connector **321**. The source connector **321** shares the same voltage with the semiconductor substrate **301**, and the source connector **323** shares the same voltage with the semiconductor substrate **300**.

[0189] In this embodiment, the source connector **323** is grounded, and voltages of the source connector **321** and the source connector **323** can reversely biased the pn junction **302** between the semiconductor substrate **300** and the semiconductor substrate **301**.

[0190] For example, the semiconductor substrate **300** has p-type dopants, and the semiconductor substrate **301** has n-type dopants. When the HEMT structure on the semiconductor substrate **301** is opened, the source connector **321** and the interconnection **381** will have high voltage, and the pn junction **302** is reversely biased. When the HEMT structure on the region **3***a* of semiconductor substrate **300** is opened, the drain connector **324** and the interconnection **381** will have high voltage, and the pn junction **302** is reversely biased. When the pn junction **302** is reversely biased, a depletion region is formed in the substrate structure **30**, and the substrate voltage of the semiconductor substrate **301** is isolated from the substrate voltage of the region **3***a* of the semiconductor substrate **300**.

[0191] In one aspect, the interconnection **381** passes through an interface formed between the nitride-based heterostructure **31** and the semiconductor substrate **301** to have an end portion embedded in the semiconductor substrate **301**. Therefore, the interconnection **381** can provide a proper electrical connection between the semiconductor substrate **301** and the source connector **321**.

[0192] In one aspect, a projection of the semiconductor substrate **301** on a top surface of the substrate structure **30** is rectangular in shape. Therefore, the semiconductor substrate **301** can carry the connectors **320-322** that form a HEMT structure.

[0193] FIGS. **37-45** are cross-sectional views of steps of a manufacturing method of the nitride-based semiconductor circuit **3**A. Referring to FIGS. **37** and **38**, the manufacturing method of the nitride-based semiconductor circuit **3**A includes: providing a substrate structure **30**. The substrate structure **30** comprises the semiconductor substrate **300** having dopants, and the semiconductor substrate has the region **3***a* and the region **3***b* being adjacent to the region **3***a*. The semiconductor substrate **301** is embedded in the region **3***b* of the semiconductor substrate **300**. The top surface **3000** of the semiconductor substrate **300** and the top surface **3010** of the semiconductor substrate **301** are coplanar. The semiconductor substrate **301** has dopants that are different from the dopants of the semiconductor substrate **300**, and the pn junction **302** is formed between the semiconductor substrate **300** and the semiconductor substrate **301**.

[0194] Referring to FIG. **39**, the manufacturing method of nitride-based semiconductor circuit **3**A includes: forming the nitride-based heterostructure **31** on the substrate structure **30**. The nitride-based heterostructure **31** is disposed on the substrate structure **30**, and the interface between the nitride-based semiconductor layers **310**, **311** and the top surface of the substrate structure **30** are parallel.

[0195] Referring to FIGS. **40-43**, the manufacturing method of nitride-based semiconductor circuit **3**A includes: forming the connectors **32** on the nitride-based heterostructure **31**.

[0196] Referring to FIG. **44**, the manufacturing method of nitride-based semiconductor circuit **3**A includes: etching through the nitride-based heterostructure **31**.

[0197] Referring to FIG. **45**, the manufacturing method of nitride-based semiconductor circuit **3**A includes: forming the connecting vias **38**. The connecting vias **38** includes interconnection **380** and interconnection **381**. The interconnection **380** electrically connects the region **3***a* of the semiconductor substrate **300** to one of the connectors **32**, and the interconnection **381** electrically connects the semiconductor substrate **301** to another one of the connectors **32**.

[0198] To be specific, referring to FIG. **37**, in the manufacturing method of the embodiment, the semiconductor substrate **300** is provided, and the semiconductor substrate has dopants. Referring to FIG. **38**, the semiconductor substrate **301** is formed through ion implantation and diffusion at high temperature. In other words, the step of providing the substrate structure includes: providing a semiconductor substrate having the dopants; and doping the other type of dopants to form the semiconductor substrate **301**, and the rest of the semiconductor substrate form the semiconductor substrate **300**. For example, the semiconductor substrate **300** has p-type dopants, and the semiconductor substrate **301** is formed through n-type ion implantation. Therefore, the top surface **3000** of the semiconductor substrate **300** and the top surface **3010** of the semiconductor substrate **301** are coplanar.

[0199] In some embodiments, the semiconductor substrate **301** may be a n-type well in the semiconductor substrate **300**.
[0200] Referring to FIG. **39**, the nitride-based semiconductor layer **311** and the nitride-based semiconductor layer **310** are formed through epitaxial growth techniques.
[0201] Referring to FIG. **40**, the gate connectors **322**, **325** are disposed on the nitride-based heterostructure **31**. The gate connector **322** includes a doped nitride-based semiconductor layer **3222** and an electrode **3221**, and the gate connector **325** includes a doped nitride-based semiconductor layer **3252** and an electrode **3251**. The doped nitride-based semiconductor layers **3222**, **3252** are disposed on the nitride-based heterostructure **31**. The electrode **3221** is disposed on the doped nitride-based semiconductor layer **3222**, and the electrode **3251** is disposed on the doped nitride-based semiconductor layer **3252**. For example, the doped nitride-based semiconductor layers **3222**, **3252** are formed through epitaxial growth techniques, and the doped nitride-based semiconductor layers **3222**, **3252** are doped with p-type dopants. Therefore, the HEMT structures as shown in FIG. **36** are normally-on HEMT structures.
[0202] Referring to FIG. **41**, the passivation layer **33** is disposed on the gate connectors **322**, **325**, and the nitride-based heterostructure **31**. For example, the passivation layer **33** includes nitride, and the passivation layer **33** covers the top surfaces and side surfaces of the gate connectors **322**, **325**.
[0203] Referring to FIG. **42**, the passivation layer **33** is etched and a plurality of openings **330** are formed. The openings **330** are for ohmic contact.
[0204] Referring to FIG. **43**, the drain connectors **320**, **324** and the source connectors **321**, **323** are disposed in the openings **330** as shown in FIG. **42**.
[0205] Referring to FIG. **44**, an ILD layer **391**, vias **36**, the patterned conductive layer **34**, an IMD layer **392**, and vias **37** are disposed above the connectors **32**, and the passivation layer **33**. The ILD layer **391** is disposed on the connectors **32** and the passivation layer **33**, and the vias **36** pass through the ILD layer **391**. The patterned conductive layer **34** is disposed on the ILD layer **391**, and the vias **36** electrically connect the patterned conductive layer **34** to the connectors **32**.
[0206] The IMD layer **392** is disposed on the patterned conductive layer **34**, and the vias **37** pass through the IMD layer **392**. The vias **37** are electrically connected to the patterned conductive layer **34**.
[0207] Also, the IMD layer **392**, the ILD layer **391**, the passivation layer **33**, the nitride-based heterostructure **31**, and part of the semiconductor substrate **301** are etched, and the openings **382** are formed. The openings **382** expose the semiconductor substrate **301**.
[0208] Referring to FIG. **45**, the patterned conductive layer **35** and the interconnections **380**, **381** are disposed. The patterned conductive layer **35** is disposed on the IMD layer **392** and the vias **37**, and the interconnections **381** are disposed in the openings **382** as shown in FIG. **44**. The patterned conductive layer **35** is located above the connectors **32**, and the patterned conductive layer **35** electrically connects to the connectors **32**. The patterned conductive layer **35** has a plurality of pads, and the pad **350** electrically connects the source connector **321** and the drain connector **324**. Also, the patterned conductive layer **34** electrically connects the connectors **32** to the patterned conductive layer **35**.
[0209] Furthermore, the passivation layer **393** as shown in FIG. **36** is disposed and covers part of the patterned conductive layer **35**, and the rest of the patterned conductive layer **35** is exposed for further connection.
[0210] The vias **36**, **37**, the patterned conductive layers **34**, **35**, and the interconnection **381** electrically connect the semiconductor substrate **301** to the source connector **321**, and the vias **36**, **37**, the patterned conductive layers **34**, **35**, and the interconnection **380** electrically connect the semiconductor substrate **300** to the source connector **323**. Therefore, during the operation of the HEMT structures of the nitride-based semiconductor circuit **3**A, the pn junction **302** is reversely biased, and the voltages of the semiconductor substrate **301** and the semiconductor substrate **300** can be different.

[0211] FIG. **46** is a top view of a nitride-based semiconductor circuit **3**B according to an embodiment of the present disclosure, and FIG. **47** is a cross-sectional view of the nitride-based semiconductor circuit **3**B according to cutting plane line I**5** in FIG. **46**, and parts of layers of the nitride-based semiconductor circuit **3**B are omitted in the top view for clarity. FIG. **46** only shows top view of a substrate structure **30**, and a patterned conductive layer **35**.
[0212] The nitride-based semiconductor circuit **3**B includes the substrate structure **30**, a nitride-based heterostructure **31**, connectors **32**, patterned conductive layers **34**, **35**, and connecting vias **38**, which is similar to the nitride-based semiconductor circuit **3**A, and the detailed description regarding similar components will not be repeated.
[0213] In this embodiment, the passivation layers **33** include oxide layer **332** and nitride layer **331**. The oxide layer **332** covers the gate connector **325**, and no oxide layer **332** is covering the gate connector **322**. The nitride layer **331** covers the oxide layer **332** on the gate connector **325**, and the nitride layer **331** directly covers the gate connector **322**.
[0214] For example, in this embodiment, the oxide layer **332** comprises silicon dioxide, and the oxide layer **332** is applied through PECVD. The nitride layer **331** comprises silicon nitride, and the nitride layer **331** is applied through LPCVD. The hydrogen concentration of oxide layer **332** is different from the hydrogen concentration of nitride layer **331**, and the doped nitride-based semiconductor layer **3252** is in direct contact with the oxide layer **332**. While the doped nitride-based semiconductor layers **3222**, **3252** are doped with p-type dopant, activation of magnesium in doped nitride-based semiconductor layer **3222** and activation of magnesium in doped nitride-based semiconductor layer **3252** can be different during high temperature thermal treatment. Therefore, the threshold voltage of the HEMT structure above the region **3***b* is different from the threshold voltage of the HEMT structure above the region **3***a*. Therefore, the HEMT structures can be prevented from accidentally being opened by a voltage spike or voltage leakage.
[0215] In one aspect, a projection of the semiconductor substrate **301** on a top surface of the substrate structure **30** is round in shape. Therefore, the semiconductor substrate **301** can provide sufficient area for deposition of the HEMT structure.
[0216] In this embodiment, the HEMT structure above the semiconductor substrate **301** is located at the high side, and the HEMT structure above the region **3***a* of the semiconductor substrate **300** is located at the low side, and the semiconductor substrate **301** is electrically connected to the source connector **321** of the HEMT structure located at the high side, and the semiconductor substrate **300** is electrically connected to the source connector **323** of the HEMT structure located at the low side. However, the present disclosure is not limited thereto.
[0217] FIG. **48** is a top view of a nitride-based semiconductor circuit **3**C according to an embodiment of the present disclosure, and FIG. **49** is a cross-sectional view of the nitride-based semiconductor circuit **3**C according to cutting plane line I**6** in FIG. **48**, and parts of layers of the nitride-based semiconductor circuit **3**C are omitted in the top view for clarity. FIG. **48** only shows top view of a substrate structure **30**, and a patterned conductive layer **35**.
[0218] The nitride-based semiconductor circuit **3**C includes a nitride-based heterostructure **31**, connectors **32**, passivation layer **33**, vias **36**, **37**, patterned conductive layers **34**, **35**, which is similar to the nitride-based semiconductor circuit **3**A, and the detailed description regarding similar components will not be repeated.
[0219] In this embodiment, the nitride-based semiconductor circuit **3**C includes a connecting vias **38**. The connecting vias **38** include an interconnection **380** and an interconnection **381**. The interconnection **380** electrically connects the region **3***a* of the semiconductor substrate **300** to a source connector **321** of the connectors **32**. The interconnection **381** electrically connects the semiconductor substrate **301** to a source connector **323** of the connectors **32**. In other words, the source connector **321** shares the same voltage with the semiconductor substrate **300**, and the source connector **323** shares the same voltage with the semiconductor substrate **301**. Furthermore, the interconnection **380** electrically connects the region **3***a* of the semiconductor substrate **300** to a

drain connector **324** of the connectors **32**.

[0220] In this embodiment, the nitride-based semiconductor circuit **3**C can form a half bridge circuit, and a drain connector **320**, a gate connector **322**, the source connector **321** and a part of the nitride-based heterostructure **31** above the region **3***a* form a HEMT structure, and the HEMT structure is located at the high side of the nitride-based semiconductor circuit **3**C. In other words, the semiconductor substrate **300** is electrically connected to the source connector **321** of the HEMT structure at the high side of the nitride-based semiconductor circuit **3**C.

[0221] The drain connector **324**, a gate connector **325**, and the source connector **323** and another part of the nitride-based heterostructure **31** above the region **3***b* form another HEMT structure, and the HEMT structure is located at the low side of the nitride-based semiconductor circuit **3**C. In other words, the semiconductor substrate **301** is electrically connected to the source connector **323** of the HEMT structure at the low side of the nitride-based semiconductor circuit **3**C.

[0222] For example, the semiconductor substrate **300** has n type dopants, and the semiconductor substrate **301** has p type dopants. The source connector **323** is grounded, while the source connector **321** will receive high voltage. Therefore, the pn junction **302** can be reversely biased during the operation of the HEMT structures above the substrate structure **30**.

[0223] FIG. **50** is a cross-sectional view of a nitride-based semiconductor circuit according to some embodiments of the present disclosure. The nitride-based semiconductor circuit **3**D has a substrate structure **30**, a nitride-based heterostructure **31**, connectors **32**, an ILD layer **391**, an IMD layer **392**, patterned conductive layers **34**, **35**, and connecting vias **38**, which are similar to the nitride-based semiconductor circuit **3**C, and the detailed description about the similar components will not be repeated.

[0224] In this embodiment, the nitride-based semiconductor circuit **3**D includes passivation layers **33**, and the passivation layers **33** include an oxide layer **332** and a nitride layer **331**. The oxide layer **332** directly touches the gate connector **325**, and the gate connector **322** is isolated from the oxide layer **332**, and the nitride layer **331** covers the oxide layer **332**.

[0225] Moreover, the gate connector **325** has an electrode **3251** and a doped nitride-based semiconductor layer **3252**, and the gate connector **322** has an electrode **3221** and a doped nitride-based semiconductor layer **3222**. The doped nitride-based semiconductor layers **3222**, **3252** are disposed on the nitride-based heterostructure **31**, and the electrodes **3221**, **3251** are disposed on the doped nitride-based semiconductor layers **3222**, **3252**, respectively. In this embodiment, the oxide layer **332** directly contacts the doped nitride-based semiconductor layer **3252**, and the electrode **3251** is isolated from the oxide layer **332**. Therefore, the threshold voltage of the HEMT structure formed by the connectors **323˜325** can be different from the threshold voltage of the HEMT structure formed by the connectors **320-322** without increasing the thickness of the nitride-based semiconductor circuit **3**D.

[0226] In an embodiment, the oxide layer **332** as shown in FIG. **50** can be applied to the nitride-based semiconductor circuit **3**B as shown in FIG. **47**.

[0227] The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical application, thereby enabling others skilled in the art to understand the disclosure for various embodiments and with various modifications that are suited to the particular use contemplated.

[0228] As used herein and not otherwise defined, the terms "substantially," "substantial," "approximately" and "about" are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can encompass instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can encompass a range of variation of less than or equal to ±10% of that numerical value, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. The term "substantially coplanar" can refer to two surfaces within micrometers of lying along a same plane, such as within 40 μm, within 30 μm, within 20 μm, within 10 μm, or within 1 μm of lying along the same plane.

[0229] As used herein, the singular terms "a," "an," and "the" may include plural referents unless the context clearly dictates otherwise. In the description of some embodiments, a component provided "on" or "over" another component can encompass cases where the former component is directly on (e.g., in physical contact with) the latter component, as well as cases where one or more intervening components are located between the former component and the latter component.

[0230] While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations are not limiting. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not necessarily be drawn to scale. There may be distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. Further, it is understood that actual devices and layers may deviate from the rectangular layer depictions of the FIGS. and may include angles surfaces or edges, rounded corners, etc. due to manufacturing processes such as conformal deposition, etching, etc. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and the drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations.

Claims

1. A nitride-based semiconductor circuit comprising: a substrate structure comprising: a first type semiconductor substrate having a first region and a second region adjacent to the first region; and a second type semiconductor substrate embedded in the second region of the first type semiconductor substrate, wherein a top surface of the first type semiconductor substrate and a top surface of the second type semiconductor substrate are substantially coplanar, and the first type semiconductor substrate has first dopants, and the second type semiconductor substrate has second dopants to form a pn junction between the first type semiconductor substrate and the second type semiconductor substrate; a nitride-based heterostructure disposed on the substrate structure, a plurality of connectors disposed on the nitride-based heterostructure; and a plurality of connecting vias going through the nitride-based heterostructure, wherein the connecting vias comprise: a first interconnection electrically connecting the first region of the first type semiconductor substrate to one of the connectors; and a second interconnection electrically connecting the second type semiconductor substrate to another one of the connectors.

2. The nitride-based semiconductor circuit of claim 1, wherein an interface between the first type semiconductor substrate and the second type semiconductor substrate is concave or concave-like from a cross-sectional perspective.

3. The nitride-based semiconductor circuit of claim 1, wherein the connectors comprise a first and second source connectors, a first and second drain connectors, and a first and second gate connectors, wherein the first gate connector is located between the first source connector and the first drain connector, and the second gate connector is located between the second source connector

and the second drain connector.
4. The nitride-based semiconductor circuit of claim 3 further comprising a second patterned conductive layer disposed on the connectors, wherein the second patterned conductive layer electrically connects to the connectors, and the second patterned conductive layer has a plurality of pads, and one of the pads electrically connects to the first source connector and the second drain connector.
5. The nitride-based semiconductor circuit of claim 4 further comprising a first patterned conductive layer disposed between the connectors and the second patterned conductive layer, wherein the first patterned conductive layer electrically connects the connectors to the second patterned conductive layer.
6. The nitride-based semiconductor circuit of claim 4, wherein the pad electrically connected to the first source connector and the second drain connector overlays part of the pn junction and electrically connects to the second type semiconductor substrate.
7. The nitride-based semiconductor circuit of claim 4, wherein the pad electrically connected to the first source connector and the second drain connector overlays both the first region and the second region.
8. The nitride-based semiconductor circuit of claim 3, wherein the first interconnection electrically connects the first type semiconductor substrate to the second source connector, and the second interconnection electrically connects the second type semiconductor substrate to the first source connector.
9. The nitride-based semiconductor circuit of claim 3, wherein the first source connector, the first drain connector, and the first gate connector are disposed on the second type semiconductor substrate, and the second source connector, the second drain connector, and the second gate connector are disposed on the first type semiconductor substrate.
10. The nitride-based semiconductor circuit of claim 3, further comprising: an oxide layer in contact with the second gate connector and isolated from the first gate connector; and a nitride layer in contact with the first gate connector.
11. The nitride-based semiconductor circuit of claim 10, wherein the second gate connector comprises: a doped nitride-based semiconductor layer disposed on the nitride-based heterostructure and in direct contact with the oxide layer; and an electrode disposed on the doped nitride-based semiconductor layer.
12. The nitride-based semiconductor circuit of claim 11, wherein the electrode is isolated from the oxide layer.
13. The nitride-based semiconductor circuit of claim 1, wherein a projection of the second type semiconductor substrate on a top surface of the substrate structure is rectangle in shape.
14. The nitride-based semiconductor circuit of claim 1, wherein a projection of the second type semiconductor substrate on a top surface of the substrate structure is round in shape.
15. The nitride-based semiconductor circuit of claim 1, wherein the second interconnection passes through an interface formed between the nitride-based heterostructure and the second type semiconductor substrate to have an end portion embedded in the second type semiconductor substrate.
16. A manufacturing method of a nitride-based semiconductor circuit comprising: providing a substrate structure, wherein the substrate structure comprises a first type semiconductor substrate having first dopants, and the first type semiconductor substrate has a first region and a second region adjacent to the first region; and a second type semiconductor substrate embedded in the second region of the first type semiconductor substrate, wherein a top surface of the first type semiconductor substrate and a top surface of the second type semiconductor substrate are coplanar, and the second type semiconductor substrate has second dopants to form a pn junction between the first type semiconductor substrate and the second type semiconductor substrate; forming a nitride-based heterostructure on the substrate structure; forming a plurality of connectors on the nitride-

based heterostructure; etching through the nitride-based heterostructure; and forming a plurality of connecting vias; wherein the connecting vias comprises: a first interconnection electrically connecting the first region of the first type semiconductor substrate to one of the connectors; and a second interconnection electrically connecting the second type semiconductor substrate to another one of the connectors.

17. The manufacturing method of claim 16, the step of providing the substrate structure includes: providing a semiconductor substrate having the first dopants; and doping the second dopants to form the second type semiconductor substrate, wherein the rest of the semiconductor substrate form the first type semiconductor substrate.

18. The manufacturing method of claim 16, wherein the connectors comprise a first and second source connectors, a first and second drain connectors, and a first and second gate connectors, wherein the first gate connector is located between the first source connector and the first drain connector, and the second gate connector is located between the second source connector and the second drain connector.

19. The manufacturing method of claim 18 further comprising forming a second patterned conductive layer disposed on the connectors, wherein the second patterned conductive layer electrically connects to the connectors, and the second patterned conductive layer has a plurality of pads, and one of the pads electrically connects to the first source connector and the second drain connector.

20. The manufacturing method of claim 19, before the step of forming the second patterned conductive layer, further comprising forming a first patterned conductive layer on the connectors, wherein the first patterned conductive layer electrically connects the connectors to the second patterned conductive layer.

21-25. (canceled)